

Can large language models predict solar irradiance in the long term?

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HIGHLIGHTS

- First approach to very long-term Global Horizontal Irradiance (GHI) prediction based on Large Language Models (LLMs).
- Proposed approach evaluated using 40 different cities located in several different types of climates.
- Comprehensive comparison with modern competitors, such as Transformers and advanced variations.
- The longest GHI prediction horizon when compared to competing methods.
- The lowest computational cost in training time per trainable parameter.

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ABSTRACT

This study introduces a novel approach for long-term Global Horizontal Irradiance (GHI) prediction using a Large Language Model (LLM), the GPT-Neo. Unlike conventional forecasting systems that rely on extensive meteorological variables or complex Transformer architectures, the proposed method operates solely on GHI time-series data and follows an enhanced pattern-recognition processing chain. A comprehensive survey of contemporary forecasting techniques highlights existing methodological gaps, particularly regarding long-term prediction. Experimental evaluations conducted using NSRDB data across diverse climatic regions demonstrate that GPT-Neo has promising predictive capability. A large-scale assessment spanning 40 locations further confirms its robustness and generalization, with GPT-Neo yielding the lowest average RMSE and the highest average R^2 among all tested models. These findings position GPT-Neo as a competitive alternative to strong baseline techniques and complex Transformer variants.

1. Introduction

Solar irradiance forecasting is a technological resource essential for enhancing the performance of photovoltaic (PV) power generation [1,2], improving reliability [3–5], efficiency [6], planning [7], and operation [8] of PV systems. More specifically, solar data prediction based on GHI has proven to be a key factor in PV systems, given that GHI is abundant [3] and widely used in installations ranging from microgrids [9] to solar farms [10]. GHI is the main component of PV energy generation because it quantifies the total amount of solar radiation received on a horizontal surface [11].

Over the years, the scientific community has developed a variety of strategies for GHI prediction. However, the most advanced methods currently employ AI-based models. Researchers frequently use models such as Long Short-Term Memory (LSTM) in many studies due to

their ability to capture temporal dependencies [12–14]. Other strategies apply Convolutional Neural Networks (CNNs) [15,16]. Besides, hybrid architectures that combine LSTM with Convolutional Neural Networks (CNN-LSTM) emerge in multimodal approaches [14,17]. However, a recent and increasingly prominent trend in the field is the adoption of Transformer-based architectures [18], which have demonstrated superior capabilities in modeling long-range temporal dependencies and managing heterogeneous data sources. Several specialized variants exemplify this evolution: Informer [19], Autoformer [20], Patch Time Series Transformer (PatchTST) [21], along with an additional set of derivative models, are featured in the study [22], reinforcing the ongoing pursuit of architectures capable of addressing incomplete data scenarios and complex time series. Similarly, the multimodal studies [17] incorporate variants such as Informer and other Transformer-based architectures within deep fusion pipelines. In [23], the use of

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the PatchTST highlights the growing maturity of these models in handling high-resolution time series combined with image data. Studies such as [13,17], and [24] also include conventional Transformer models, demonstrating that both modern variants and classical architectures are increasingly relevant in the current landscape of GHI forecasting. However, several gaps remain unaddressed, such as developing models that can incorporate contextual information into the data (as achieved by Large Language Models (LLMs)) and extending the forecasting horizon.

One crucial point is the forecasting horizon, or the so-called horizon of prediction. The forecasting horizon is the number of steps into the future that each approach predicts, i.e., the GHI values for the next 24 hours, $\text{horizon} = 24$. The literature presents several works regarding GHI prediction, and each work uses a specific size for the forecasting horizon. Researchers define the size according to the purpose of the application. According to the review in [25], there are four well-defined categories: very short term, short term, medium term, and long term. Commonly used for real-time monitoring, the very short-term horizon ranges from seconds to minutes. Applied for unit commitment, the short-term forecasting runs from a short period of hours to a long period of hours, a few days. Employed for maintenance programming, the medium-term horizon uses week periods. Focused on operations management, the long-term forecasting is associated with months and years. Here, it is important to highlight that one of the motivations of the present manuscript is the long-term prediction associated with LLMs. One of our main research investigations in this manuscript is: Are LLMs ready to shoulder GHI long-term prediction? Few works handle GHI forecasting based on LLMs. Furthermore, it is worth noting that additional variables can physically affect solar irradiance. At the same time, we emphasize that the objective of this study is not to propose a complete physical model of solar irradiance, but rather to evaluate a data-driven method with low computational demand based on the GHI time series. By using a GHI univariate series, the pipeline reduces the volume of input data, the preprocessing effort, memory usage, and computational demand. This requirement simplifies the training and deployment of advanced models on more affordable hardware platforms, especially in scenarios where high-performance computing resources or comprehensive meteorological datasets are not readily available. In addition, reducing data volume can lower energy costs associated with training models, particularly given the current intensive use of data centers. Finally, it is important to mention that the proposed system is currently evolving as part of a research project that brings together Western University (Canada) and the University of Campinas (UNICAMP), Brazil, in a binational collaboration.

Main contributions

- **Novel approach for long-term GHI prediction:** Our work proposes a pioneering system for long-term GHI forecasting based on a large language model (LLM). The proposed system follows an pattern recognition processing chain based on GHI data and LLM architecture, the GPT-Neo. The evaluation metrics, used to assess the results, suggest that GPT-Neo is a competitive alternative to Transformers and their advanced variants currently available. Additionally, it is important to note that improvements are also attributed to the model's capacity, given that it is a modern LLM architecture. In line with this original contribution, the paper also presents a novel analysis of the GPT-Neo architecture applied to long-term GHI forecasting using data collected from the National Solar Radiation Database (NSRDB) under realistic weather conditions [26,27].
- **Practical implications:** GPT-Neo outperforms all competing models across key metrics for long-term GHI forecasting in Recife. It achieves the lowest MAE (12.41), RMSE (14.91), and JSD (0.0105), while also attaining the highest R^2 (0.7293), indicating superior robustness among the others. These results highlight the potential

of LLM-based models over traditional Transformer variants for tropical climate prediction (a bad case for all models). In Reno, GPT-Neo outperforms all evaluated models, achieving the lowest MAE (6.69), RMSE (8.38), and JSD (0.0058), and the highest R^2 (0.9505). These results confirm GPT-Neo's superiority and stability in long-term GHI prediction under Mediterranean-like conditions (a good case for all models). Across 40 diverse locations, GPT-Neo demonstrates robust overall performance among all models, achieving the lowest average RMSE (11.81) and the highest R^2 (0.7787), indicating both precision and strong variance explanation. These findings confirm the effectiveness of LLM-based forecasting, especially GPT-Neo, as a robust and generalizable solution for large-scale climate prediction.

- **Advances in the field of long-term GHI prediction:** Our study offers significant contributions to key aspects of long-term GHI forecasting by proposing a comprehensive and robust system that encompasses all stages, from data preprocessing to the forecasting core. It employs advanced machine learning architectures that are highly robust and have reduced implementation complexity, relying solely on GHI time series. Furthermore, the study addresses existing gaps in current methodologies by enhancing long-term GHI forecasting using the large language model GPT-Neo, thereby improving overall system performance. Notably, the proposed approach establishes a new benchmark for long-term GHI forecasting methods based on LLMs. This advancement can foster the development of more robust, accurate, and reliable systems, generating a positive impact in both academic research and, especially, in practical applications across the renewable energy industry, with particular relevance to photovoltaic systems.
- **A comprehensive survey of modern approaches:** We present a rigorous literature review of current methods, focusing on essential components required for a robust GHI prediction system. These include the implementation of system-specific modules, the selection of input variables, preprocessing techniques, feature engineering strategies, the types of regression models employed, and the robustness and forecasting horizon of competing systems. This analysis clarifies recent advancements in long-term GHI prediction technologies, emphasizing the critical factors that shape both performance and practical applicability.

2. Related works

The literature on solar irradiance forecasting and photovoltaic power generation exhibits substantial methodological diversity, both in data usage and modeling techniques. In recent years, a clear trend toward AI-based models has emerged, with considerable variation in methodological rigor and data-processing practices. In [12], the authors investigate multi-step forecasting of global irradiance using an LSTM model, yet without performing data cleaning or transformation, nor employing established methods for variable selection or hyperparameter tuning, relying instead on a limited set of radiometric and meteorological variables. In contrast, the study presented in [28] applies structured preprocessing procedures (including removal of nighttime data, filtering based on solar elevation angle, and normalization) alongside feature engineering grounded in clearness and clear-sky indices. This work also employs mutual information for variable selection and compares multiple models, including Multi-Layer Perceptron (MLP), Random Forest, and Scaled Persistence; however, [12,28] impose a restricted prediction horizon.

In [29], the researchers broadened the scope by integrating photovoltaic power variables with meteorological data in Nonlinear Autoregressive Exogenous (NARX), ANN, Support Vector Machine for Regression (SVR), and Gaussian Process Regression (GPR) models. Nevertheless, they provide insufficient detail on preprocessing procedures, employ unspecified normalization techniques, and rely on trial-and-error hyperparameter tuning. The reference [15] presents a

similar approach, where the authors remove outliers and impute missing data, a practice that may introduce bias. Regardless, they do not adopt any established variable selection method. In this study, the authors compare models such as Echo State Network (ESN) and CNN for hourly forecasting, once again restricting their analysis to short-term prediction horizons.

The progression toward modern architectures is evident in [13], which proposes a Transformer model for photovoltaic forecasting using meteorological variables and Numerical Weather Prediction (NWP) data. Nevertheless, the study lacks robust preprocessing and employs simple Pearson correlation for feature selection, which, although valid, presents limitations when dealing with nonlinear relationships. In [30], the authors introduce the U-LSTM-AFT model, integrating U-Net, LSTM, and Attentional Feature Transformation (AFT), with a more consistent preprocessing procedure based on outlier removal and Min–Max normalization. However, the study does not incorporate formal mechanisms for variable selection or hyperparameter optimization. In [16], the researchers explore horizontal federated learning for irradiance forecasting by combining CNNs, MLPs, LSTMs, and SA-CNN-LSTMs. Although the system removes anomalous samples and fills missing data, the manuscript fails to describe the methods it employs clearly. Once again, the prediction horizon remains limited.

In addition, several studies employ sky images, as seen in [14,17,23,31,32]. These works apply different levels of image preprocessing, including distortion correction, visual token extraction, and multimodal fusion, and they adopt advanced architectures such as CNN-LSTM, Swin Transformer, TimesNet, T2T-ViT, and PatchTST. Although these approaches are advanced, many do not follow systematic feature selection or hyperparameter tuning methodologies, which limits reproducibility. Finally, in [24], the authors use VMD decomposition as a feature engineering technique and apply models such as SVR, BPNN, LSTM, and TFT for day-ahead forecasting. Yet, they do not employ formal hyperparameter optimization methods.

By synthesizing these studies, we observe significant advances in modeling, particularly through the use of Transformer-based approaches. However, critical gaps remain regarding preprocessing standardization, feature engineering, variable selection, and systematic hyperparameter tuning. Most studies rely on a large number of meteorological variables, complex image-based models, or NWP data, which increase operational complexity and create challenges for real-world deployment. In addition, the prediction horizons remain extremely limited.

Finally, the comparative analysis of Tables 1–3 (Parts I, II, and III) reveals differences among the studies with respect to the main components of a robust solar forecasting system: input variables, data preprocessing, attribute preprocessing, feature engineering, regression method, forecasting horizon, and robustness. The input-variable column highlights the degree to which each study depends on different types of data, ranging from basic meteorological measurements to multimodal information and sky images, which directly influence the complexity and generalization capability of the models. Data preprocessing reflects the level of care dedicated to properly preparing datasets, including procedures such as outlier removal, gap treatment, and temporal standardization, all of which are essential for avoiding distortions and ensuring statistical consistency. Feature engineering indicates the extent of enrichment applied to the original variables, encompassing the construction of indices, decomposition techniques, and multi-modal fusion. The regression method demonstrates the variety of approaches adopted, spanning from classical models to modern architectures. Finally, the forecasting horizon differentiates the level of complexity addressed by each study, ranging from minutes to several days. Based on this structure, we classified methods that lacked one or more of the core components as weak, those that included all components but operated only within short-term horizons as intermediate, and those that included all components and achieved horizons of a few days as highly robust. The proposed method comprehensively incorporates all components and, in addition, achieves forecasting horizons extending over multiple years, making it more robust than all competing approaches and superior in both performance and scope.

3. Theoretical background

This section introduces the theoretical background related to this manuscript, such as different climates, as well as the employed models: Transformer, Informer, Autoformer, Temporal Fusion Transformer (TFT), Patch Temporal Fusion Transformer (PatchTFT), and Large Language Models (LLMs).

3.1. Climates

To deal with long-term forecasting in solar generation, we need to consider that the weather affects energy generation. Since the models are data-driven, their performance is directly affected by the quality and diversity of the input data. Moreover, we rely only on GHI time series,

Table 1
List of approaches (Part I).

Method	Input	Data Cleaning	Data Transformation	Feature Engineering	Model	Horizon	Robustness
[12]	DHI, DNI, dew point, temperature, pressure, relative humidity, wind direction, wind speed	No	No	Selects relevant variables without a validated method in the literature	LSTM (lagged model)	3h–24h ahead	Weak
[28]	GHI, DNI, DHI	Yes (removal of nighttime data and exclusion for solar elevation < 10°)	Yes (clearness index, clear-sky index, component-specific indices)	Mutual information method	Scaled persistence, MLP, Random Forest (lagged models)	1h–6h	Weak
[29]	Power, temperature, DHI, DNI	Yes (method not detailed)	Yes (normalization, type not specified)	NARX	ANN, SVR, GPR	1h–6h	Weak
[15]	Temperature, relative humidity, wind speed, wind direction, diffuse radiation, GHI, precipitation	Yes (outlier removal and missing-data imputation)	Yes (Min–Max normalization)	No	ESN, CNN	1h	Weak
[13]	PV power, solar radiation, PV module temperature, NWP solar radiation, NWP precipitation, NWP temperature, pressure, humidity, wind speed	No	No	Pearson correlation coefficient (PCC)	Transformer, ANN, LSTM	1h	Weak
[30]	GHI, wind speed, mean temperature, mean humidity	Outlier removal	Min–Max normalization	No formal mechanism	U-LSTM-AFT	1h	Weak
[16]	GHI, temperature, relative humidity	Removal of abnormal/missing samples (method not specified)	Not specified	No	CNN, MLP, LSTM, SA-CNN-LSTM	1 day	Weak

Table 2
List of approaches (Part II).

Method	Input	Data Cleaning	Data Transformation	Feature Engineering	Model	Horizon	Robustness
[22]	GHI, temperature, humidity	Removal of inconsistent/anomalous data	Z-score	Wavelet Packet Decomposition (WPD)	ID-Informer, Informer, Fedformer, Mask-Transformer, LSTM, TCN, RF, XGBoost, Bert, MARS ANN (MLP)	1h, 3h, 6h	Intermediate
[33]	Meteorological variables (T, WS, WD, cloud cover levels, solar zenith angle, DNI, GHI); aerosol optical depths at multiple wavelengths	Removal of constant-value rows	Standardization (zero mean, unit std)	Experimental variable-combination evaluation		25h, 48h, 72h	High
[14]	Time of day/year, clear-sky index, clear-sky irradiance, wind, temperature, humidity, pressure, cloud-image features, noise signal	Value clipping	Cyclic trig transforms + Min-Max normalization	No	CNN-LSTM	2h	Weak
[31]	Normal + underexposed sky images + temporal information	Removal of useless/non-representative data	No consolidated method	Multimodal data fusion	Swin Transformer, DNN, TimesNet, TSMixer, DLinear, NUT, NUTF	1h	Weak
[17]	GHI, sky images, multimodal combined data	No	No	Multimodal fusion	MMDC, MMIF, TSDC, GIDC, CLSTM, Transformer, LSTM, MLP, Ensemble NWP	15min to 24h	Weak

Table 3
List of approaches (Part III).

Method	Input	Data Cleaning	Data Transformation	Feature Engineering	Model	Horizon	Robustness
[23]	GHI, satellite images, cloud masks, clear-sky model, temp, humidity, solar angles, cloud cover index	Image-distortion removal	Normalization, standardization, SZA/SAA computation	Correlation + p-value + Lasso + feature extraction from GSI	LSTM, PatchTST	1h	Intermediate
[32]	GHI, solar zenith angle, humidity, cloud cover, wind, precipitation, snow depth, temperature, sky images	Sky-image distortion correction	Time-series to 2D tensor + image unfolding into tokens	No	LSTM, Informer, Autoformer, DLinear, LightTS, PatchTST, ResNet, ViT, T2T-ViT	10min	Weak
[24]	GHI, ambient temperature, relative humidity (when available), NWP GHI	No	No	Variational Mode Decomposition (VMD)	SVR, BPNN, LSTM, TFT, Encoder-Decoder	1 day	Weak
Proposed System	GHI (optimized, low-complexity variable set)	Yes (outlier removal and sync checking)	Yes (Min-Max)	Quantized GHI	Transformers, Variations of Transformers and the GPT-Neo)	Adjustable (Very Long-Term)	Superior

which means that we do not use other climate parameters as sources of information to feed the models, making them more sensitive to the limitations of a single source of information. Hence, due to the climatic uncertainties involved, it is vital to apply data from different locations, ensuring global coverage across diverse climate zones. Aware of this situation, we adopted the Köppen-Geiger climate classification to select locations throughout the Earth [34–36].

The Köppen-Geiger climate classification, as summarized in Table 4, organizes the climate zones based on the main climate, temperature and precipitation patterns. The classification represents each climate through a two- or three-letter code, where the first letter indicates the principal climate type (tropical, arid, temperate, etc.), the second describes the rainfall regime (“f” for fully humid, “s” for dry summer, “m” for monsoon, etc.), and the third, when included, indicates the thermal characteristic of the summer or year (“a” for hot summer, “b” for mild summer, “c” for cold summer, etc.).

3.2. Transformer

The Transformer was proposed in 2017 [18] and introduced a new paradigm in Artificial Intelligence. Instead of recurrent or convolutional layers, the Transformer structure uses only attention mechanisms. Attention mechanisms can capture the global dependencies between input and output. Initially, the authors applied Transformers to language translation; however, researchers around the world have expanded

Table 4
Köppen-Geiger climate classification [34–36].

Code	Main climate	Precipitation criterion	Temperature criterion
Af	Tropical	Fully humid (f)	—
Am	Tropical	Monsoon (m)	—
As	Tropical	Dry summer (s)	—
BSh	Arid	Steppe (S)	Hot (h)
BSk	Arid	Steppe (S)	Cold (k)
BWh	Arid	Desert (W)	Hot (h)
BWk	Arid	Desert (W)	Cold (k)
Cfa	Temperate	Fully humid (f)	Hot summer (a)
Cfb	Temperate	Fully humid (f)	Warm summer (b)
Cfc	Temperate	Fully humid (f)	Cold summer (c)
Csa	Temperate	Dry summer (s)	Hot summer (a)
Csb	Temperate	Dry summer (s)	Warm summer (b)
Cwb	Temperate	Dry winter (w)	Warm summer (b)
Dfa	Continental	Fully humid (f)	Hot summer (a)
Dfb	Continental	Fully humid (f)	Warm summer (b)

the Transformer application to many different areas (natural language processing, time series prediction, and computer vision), and numerous Transformer-based architectures have been developed, including Informer [19], Autoformer [20], TFT [37], PatchTFT [21], and ViT [38]. Besides, it is relevant to highlight that the Transformer serves as the foundation for Large Language Models, and the GPTs exist in recent years due to the Transformer architecture.

3.3. Informer

The Informer architecture was introduced in 2021 [19]. The authors proposed this model to address Long-Sequence Time-Series Forecasting (LSTF). One of the main goals was to efficiently increase the capacity for capturing the dependencies of the pair output-input involved in LSTF. According to the authors, the Informer has three principal advantages: (i) reduced time complexity in memory usage; (ii) due to the proposed structure, the Informer accentuates the dominant attention and efficiently operates on extremely long input sequences; and (iii) improves the inference speed in long sequences because it uses only one forward operation in long time-series prediction.

3.4. Autoformer

The Autoformer was presented in 2021 [20]. Motivated by real-world applications, the authors developed the Autoformer to address current-relevant issues, such as long-term energy planning demand and meteorological event alerts. Instead of the self-attention mechanism used in Transformers, the Autoformer structure uses a decomposition approach based on the Auto-Correlation mechanism. Among the advantages, the authors highlight (i) the computational efficiency and (ii) the robustness of the performance in diverse real-world applications.

3.5. Temporal fusion transformer (TFT)

The Temporal Fusion Transformer (TFT) was introduced in 2021 [37]. TFT combines recurrent layers—for local processing—and interpretable self-attention layers—to focus on long-term dependencies. The TFT model presents gating mechanisms that adapt the depth and complexity of the structure, skipping unused structural elements. As per the authors, TFT seeks to achieve high performance for multi-horizon prediction, and it has presented robust results for different real-world datasets (electricity, traffic, retail, and volatility databases), surpassing reference benchmarks.

3.6. Patch temporal fusion transformer (PatchTFT)

The PatchTFT was proposed in 2022 [21]. This approach segments the temporal series into patches and uses independent channels for each univariate temporal series, sharing the embedding and Transformer weights among all these series. According to the authors, among the main advantages, the PatchTFT model reduces time and space complexity, improves representation learning, and enhances the capacity to learn from longer historical windows. Nowadays, researchers apply the PatchTFT model in a wide range of real-world applications (weather, traffic, and electricity prediction).

3.7. Large language models (LLMs)

LLMs have emerged as a breakthrough in AI, particularly after the introduction of transformer-based architectures capable of learning complex patterns and capturing long-range dependencies from extensive corpora of textual data. These models are pretrained on large-scale datasets and acquire broad linguistic and semantic knowledge, enabling them to perform a wide variety of tasks, such as question answering, summarization, reasoning, code generation, and dialogue, without requiring task-specific training. One of their central strengths is the ability to generalize across domains by integrating contextual understanding and pattern recognition within a unified computational framework [39].

In recent years, researchers have been exploring LLMs for scientific and numerical domains, including time series forecasting. In this context, they reformulated temporal sequences as tokenized inputs, which enable LLMs to combine numerical information with semantic reasoning guided by prompt-based instructions. The advantages include a more natural incorporation of contextual data, improved handling of heterogeneous or sparse datasets, and the ability to generalize to new forecasting tasks with minimal retraining. Because of these characteristics, LLMs have gained increasing attention in applications such as

financial prediction, energy demand forecasting, environmental monitoring, and anomaly detection, establishing a new direction within the time series forecasting community [39].

4. Proposed approach

Fig. 1 summarizes our proposed system, depicting the main steps: a wide variety of climates and subclimates collected from NSRDB (raw data), data cleaning, resampling, quantization, normalization, LLM, and predicted GHI.

Fig. 1(a) details the initial phase of the scientific data pipeline focused on the analysis, monitoring, and prediction of renewable energy around the globe. The process begins with large-scale geospatial screening, where we employ a wide variety of climates and subclimates across the earth, starting with areas of high solar incidence. From there, we apply a selection of 40 distinct geographic locations, represented as solar potential assets (photovoltaic panels) that undergo a cartographic individualization process. For each of these numbered points, we establish a communication bridge with the NSRDB, which is a high-fidelity meteorological data source widely used in the industry. More specifically, Algorithm 1 establishes the connection to the database server. Then, the algorithm configures the client with the set of geolocations $\Omega = \{(\phi_i, \lambda_i)\}_{i=1}^{40}$, where ϕ_i and λ_i represent, respectively, the latitude and longitude of each location i , as well as the time interval of interest $[t_s, t_e]$ and the sampling resolution Δt . Next, for each pair of coordinates $(\phi_i, \lambda_i) \in \Omega$, the algorithm makes an individual query to the NSRDB database to obtain the raw GHI time series $G_i(t)$, defined for $t \in [t_s, t_e]$ with resolution Δt . After data is returned, the algorithm standardizes the time records to ensure time zone consistency and uniform indexing. At the end of the first block, the system stores all series in the set $\mathcal{G} = \{G_i(t)\}_{i=1}^{40}$, resulting in a structured collection of raw GHI time series. The main objective of this integration is to acquire the GHI time series, which represents the total solar radiation. The final result of this procedure, shown on the right side of the diagram in Fig. 1(a), is the acquisition of raw GHI time series (represented by the red wave profiles).

The image detailed in Fig. 1(b) of the data processing flow focuses specifically on the treatment, refinement, and statistical standardization of GHI time series. The process begins with the receipt of raw GHI data obtained in Algorithm 1, which often presents heterogeneity in terms of sampling and integrity. To verify these inconsistencies, we submit the data to a rigorous stage, represented by Algorithm 2, which processes the raw dataset $\mathcal{G} = \{G_i(t)\}_{i=1}^{40}$ by verifying temporal coverage, detecting gaps or inconsistencies, and replacing missing values with zero to ensure structural continuity. After consolidating data integrity, the flow moves on to the resampling stage, a mathematical procedure essential for normalizing the time scale. First, high-frequency measurements are converted from an hourly resolution (solid red lines in Fig. 1(b)) to a monthly scale (dotted blue lines in Fig. 1(b)). Here, each hourly series $G_i(t)$ is resampled to a monthly resolution, producing resampled GHI series $\mathcal{G}^{(m)} = \{G_i^{(m)}(\tau)\}_{i=1}^{40}$.

Fig. 1(c) consists of data preparation and LLM execution for the generation of solar irradiance forecasts. The flow begins with monthly data from the previous phase, and now performs the quantization and technical compression process using a 128-level Analog-to-Digital Converter (ADC) with 7-bit resolution. This component transforms smooth curves into quantized signals, represented by step graphs, which simplify the complexity of the information to facilitate computational processing. Next, the resulting signals undergo a min-max normalization step, where the values are adjusted to a standard scale between 0 and 1. Both procedures (quantization and normalization) are detailed in Algorithm 3.

After quantization and normalization, as shown in Fig. 1(c), the proposed system sends the normalized data to the hardware dedicated to LLM training (the CPU and GPU server). In this phase, the proposed system executes Algorithm 4. More specifically, the system transfers each normalized time series $\hat{G}_i(t)$ to Algorithm training server and the 3 divides each one into three subsets: the training set $D_i^{(t)}$, the validation

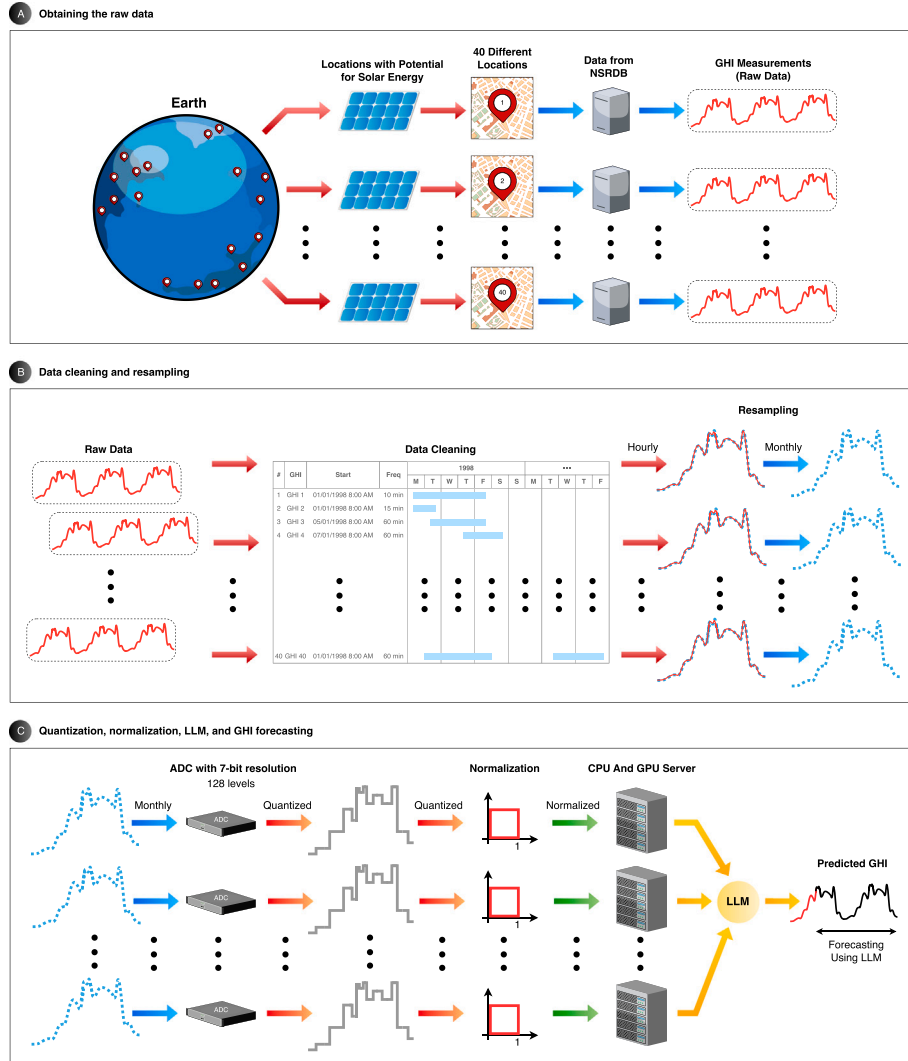


Fig. 1. This diagram comprises the proposed system, depicting the main steps: (a) obtaining the raw data, (b) data cleaning and resampling, (c) quantization, normalization, LLM, and predicted GHI.

Algorithm 1 GHI acquisition pipeline.

Require: Set of geolocations $\Omega = \{(\phi_i, \lambda_i)\}_{i=1}^{40}$ (latitude, longitude); NSRDB access (NREL API/key); temporal range $[t_s, t_e]$; sampling resolution Δt (hourly)

Ensure: Set of raw GHI time series $\mathcal{G} = \{G_i(t)\}_{i=1}^{40}$ aligned to $[t_s, t_e]$ with Δt

- 1: Initialize the NSRDB databases access through NREL API/key
- 2: Initialize NSRDB client with $\Omega = \{(\phi_i, \lambda_i)\}_{i=1}^{40}$, $[t_s, t_e]$, and Δt
- 3: Initialize storage for $\mathcal{G}$
- 4: **for all** $(\phi_i, \lambda_i) \in \Omega$ **do**
- 5: Query NSRDB for location i using (ϕ_i, λ_i) and request GHI data
- 6: Retrieve raw series $G_i(t)$ for $t \in [t_s, t_e]$ at resolution Δt
- 7: Standardize timestamps (timezone/UTC) and ensure consistent indexing
- 8: Store $G_i(t)$ as the raw GHI time series for location i
- 9: **end for**
- 10: Return $\mathcal{G} = \{G_i(t)\}_{i=1}^{40}$

Algorithm 2 GHI data integrity verification and resampling.

Require: Raw hourly GHI series $\mathcal{G} = \{G_i(t)\}_{i=1}^{40}$ over $[1998, 2018]$; resolution Δt

Ensure: Monthly standardized GHI series $\mathcal{G}^{(m)} = \{G_i^{(m)}(\tau)\}_{i=1}^{40}$

- 1: **for all** $i \in \{1, \dots, 40\}$ **do**
- 2: Verify temporal coverage and detect gaps in $G_i(t)$.
- 3: **if** Data leakage, missing values, or temporal gaps are detected in each $G_i(t)$ **then**
- 4: Discard the corresponding $G_i(t)$
- 5: Replace it with a new city containing complete/consistent data
- 6: **end if**
- 7: Resample to monthly resolution:
- 8: $G_i^{(m)}(\tau) \leftarrow \text{RESAMPLEMONTHLY}(G_i(t))$
- 9: **end for**
- 10: Store $\mathcal{G}^{(m)} = \{G_i^{(m)}(\tau)\}_{i=1}^{40}$

set $D_i^{(v)}$, and the test set $D_i^{(e)}$ (evaluation set). Next, the algorithm applies the forecast horizon H , which determines the future time interval

to be estimated. Then, the algorithm initializes the $\mathcal{M}$ model with the set of hyperparameters $\Theta = \{\mathcal{L}, \mathcal{A}, \mathcal{E}, l_r, \beta_{lr}, \beta_{va}, \zeta, n_{dr}, n_{he}, n_{hi}\}$, where $\mathcal{L}$ represents the loss function, $\mathcal{A}$ the optimizer, $\mathcal{E}$ the training epochs, l_r

Algorithm 3 GHI signal quantization and normalization.

Require: Monthly GHI time series $\mathcal{G}^{(m)} = \{G_i^{(m)}(\tau)\}_{i=1}^{40}$
Ensure: Quantized and normalized GHI series $\mathcal{G}^{(n)} = \{\hat{G}_i(\tau)\}_{i=1}^{40}$ in range $[0, 1]$

- 1: Define ADC resolution $b = 7$ bits
- 2: Compute number of quantization levels $L = 2^b = 128$
- 3: **for all** locations $i \in \{1, \dots, 40\}$ **do**
- 4: Apply ADC quantization to monthly series:
- 5: $G_i^{(q)}(\tau) \leftarrow \text{QUANTIZE}(G_i^{(m)}(\tau), L)$
- 6: Compute minimum and maximum values:
- 7: $G_{\min} \leftarrow \min(G_i^{(q)})$
- 8: $G_{\max} \leftarrow \max(G_i^{(q)})$
- 9: Apply min-max normalization:
- 10: $\hat{G}_i(\tau) \leftarrow \frac{G_i^{(q)}(\tau) - G_{\min}}{G_{\max} - G_{\min}}$
- 11: **end for**
- 12: Store $\mathcal{G}^{(n)} = \{\hat{G}_i(\tau)\}_{i=1}^{40}$

Algorithm 4 Long-term GHI forecasting using the ML model.

Require: Normalized GHI time series $\hat{G}_i(t)$; forecast horizon H ; training set $\mathcal{D}_i^{(t)}$; validation set $\mathcal{D}_i^{(v)}$; test set $\mathcal{D}_i^{(e)}$; model $\mathcal{M}$, hyperparameters $\Theta = \{\mathcal{L}, \mathcal{A}, \mathcal{E}, l_r, \beta_{lr}, \beta_{va}, \varsigma, n_{dr}, n_{he}, n_{hi}\}$
Ensure: Long-term predicted GHI series $\hat{Y}_i(t)$ over horizon H

- 1: Transfer $\hat{G}_i(t)$ to the training server (CPU/GPU environment)
- 2: Split $\hat{G}_i(t)$ into $\mathcal{D}_i^{(t)}$, $\mathcal{D}_i^{(v)}$, and $\mathcal{D}_i^{(e)}$
- 3: Initialize $\mathcal{M}$ with Θ , and H
- 4: Train $\mathcal{M}$ with $\mathcal{D}_i^{(t)}$
- 5: **for** $\epsilon \leftarrow 1$ to $\mathcal{E}$ **do**
- 6: Sample a batch of size β_{lr}
- 7: Compute model predictions, and evaluate the loss function $\mathcal{L}$ on $\mathcal{D}_i^{(t)}$ and $\mathcal{D}_i^{(v)}$
- 8: Update model parameters via $\mathcal{A}$
- 9: **end for**
- 10: Generate long-term forecast over horizon H :
- 11: $\hat{Y}_i(t) \leftarrow \text{FORECAST}(\mathcal{D}_i^{(e)}, H, \Theta)$
- 12: Return $\hat{Y}_i(t)$

the learning rate, β_{lr} represents the batch size, β_{va} the validation batch size, ς the seed, n_{dr} the dropout rate, n_{he} the number of attention heads in the reprogramming layer and n_{hi} denotes the number of hidden layers in the LLM backbone. After training is complete, the system uses the model to generate the predicted series $\hat{Y}_i(t)$ over the forecast horizon H , based on the data from $\mathcal{D}_i^{(e)}$, providing the GHI long-term prediction. The last graph in Fig. 1(c) illustrates the system's ultimate objective by presenting the red prediction curve, which shows how the LLM forecasts future solar radiation using the complete data pipeline within the proposed system.

Besides, the proposed approach reprograms the LLM to perform very long-term forecasting. As shown in Fig. 2(A), we use the philosophy proposed in [39], where Prompt-as-Prefix (PaP) directs the transformation of reprogrammed input patches. On the other hand, Fig. 2(B) shows the instruction prompts (PaP) used in this work. It is relevant to mention that according to reference [39], the prompt is part of the LLM's reprogramming/conditioning mechanism, and it provides task-level instructions regarding the prediction problem. In our context, the prompt provides minimal information for GHI prediction, including the nature of the input sequence and the desired prediction horizon. Therefore, the prompts can be interpreted as a task-level specification within the LLM learning.

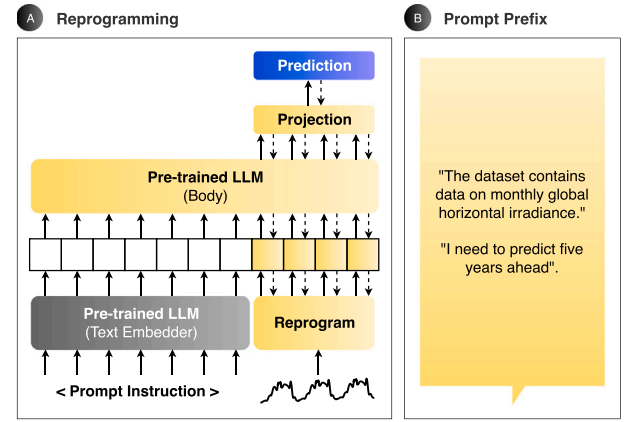


Fig. 2. LLM reprogramming (Adapted from [39]).

5. Results

In this section, we employed different metrics to ensure a holistic view of the proposed approach's performance. To achieve this, this study utilizes Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), coefficient of determination (R^2), Jensen-Shannon Divergence (JSD), and Overlap Coefficient (OVC).

MAE is a statistical metric derived from a measurement of average error [40]. It is a measure widely used to evaluate the robustness of regression models [41]. By disregarding the direction of the errors, the MAE measures the mean magnitude of the differences between the target and the forecasted values. The MAE informs, on average, how distant the predictions are from the expected values. Furthermore, it is an intuitive criterion because it has the same unit as the variable of interest. The MAE can be determined by

$$\text{MAE} = \frac{1}{N_d} \sum_{i=1}^{N_d} |y_i - \hat{y}_i|, \quad (1)$$

where N_d is the total number of data points or observations, y_i is the actual (observed) value for the i -th sample, and $\hat{y}_i$ is the predicted (estimated) value for the i -th sample, while $|\cdot|$ is the absolute value operator.

RMSE is an error measure obtained from the root of the Mean Squared Error (MSE) [42]. This metric is also widely used in regression problems [43]. The RMSE indicates, on average, how much the predictions deviate from the expected values; however, it gives more weight to larger errors because this measure employs the squaring of the differences. The RMSE also uses the same unit as the variable of interest. The RMSE can be estimated by

$$\text{RMSE} = \sqrt{\frac{1}{N_d} \sum_{i=1}^{N_d} (y_i - \hat{y}_i)^2} \quad (2)$$

R^2 , also referred to as the Coefficient of Determination, is a metric that indicates how well the model fits the observed data [44]. Due to it, the R^2 is frequently applied in regression problems to evaluate AI-models' performance [45]. An R^2 value of 1 means that the model explains 100% of the variance in the dependent variable, indicating a perfect fit of the data. On the other hand, an R^2 of 0 suggests that the model explains none of the variance in the dependent variable. R^2 can even be negative, meaning that the model performs worse than using the mean as a prediction. It is relevant mention that R^2 is dimensionless. The R^2 can be calculated by

$$R^2 = 1 - \frac{\sum_{i=1}^{N_d} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{N_d} (y_i - \bar{y})^2} \quad (3)$$

JSD is a similarity measure between two or more probability distributions [46]. The closer the JSD value is to 0, the more identical the distributions are. Based on this, JSD offers a robust and interpretable tool for quantifying differences between distributions. It is worth noting that the JSD is a dimensionless metric. The JSD can be expressed as

$$\text{JSD}(P \parallel Q) = \frac{1}{2} D_{\text{KL}}(P \parallel M) + \frac{1}{2} D_{\text{KL}}(Q \parallel M), \quad (4)$$

where P and Q are two discrete probability distributions (from expected and estimated data), $D_{\text{KL}}(\cdot)$ denotes the Kullback-Leibler divergence, and M is the average (or mixture) distribution of P and Q , representing a midpoint between them.

OVC is a metric that quantifies the degree of overlap between two probability distributions [47]. Based on the shared probability mass between the distributions, the OVC indicates how similar they are. Its value ranges from 0 to 1, where 1 means the distributions are identical, and 0 indicates they are completely disjoint (i.e., no superposition). It is important to mention that OVC is also dimensionless. The OVC can be obtained by

$$\text{OVC}(P, Q) = \sum_{i=1}^n \min(P(i), Q(i)), \quad (5)$$

where P and Q are probability distributions defined under the same discrete event space $i = 1, \dots, n$, and the function $\min(\cdot)$ refers to the minimum operator, which returns the smaller of the two values from the distributions.

It is worth noting that, in addition to MAE, RMSE, and R^2 , we included distributional metrics, such as JSD and OVC, to assess whether the forecasting models preserve the statistical behavior of the observed GHI series over a multi-year forecasting horizon, based on monthly-averaged values. The JSD metric measures the discrepancy between the empirical distribution of the observed GHI values and the distribution generated by the forecasts. Lower JSD values indicate that the predicted series better preserves the statistical profile of the measured irradiance. The OVC metric, in turn, provides a complementary interpretation by quantifying the shared probability mass between the observed and predicted distributions. Higher OVC values indicate greater overlap and, consequently, stronger distributional agreement. JSD and OVC are particularly interesting for multi-year GHI forecasting based on monthly-averaged values, since solar-planning decisions depend on the model's ability to represent the distribution of monthly average irradiance over time. In other words, in addition to reducing the average forecasting error, the model should reproduce the occurrence of months with lower, intermediate, and higher solar availability, as these variations influence annual energy-yield estimates, preliminary photovoltaic system sizing, seasonal-scale storage planning, and energy-risk assessment. Therefore, the JSD and OVC metrics provide additional evidence of the model's ability to reproduce the long-term variability structure of solar irradiance, which is valuable in a solar-energy decision-making process substantiated on a data-driven method.

To run our proposed approach, we utilized the Python language and hosted it on the Google Colab Data Science and Machine Learning tool, equipped with 53.0 GB of RAM, an Intel Xeon CPU, and an NVIDIA L4 GPU. More specifically, the Intel Xeon processor operates at a base frequency of 2.20 GHz, with a 64-bit processor architecture, where each physical core has 2 threads (Hyper-Threading technology), resulting in a total of 12 logical cores (6 physical $\times$ 2 threads/physical). The system is running within a virtual machine managed by the Kernel-based Virtual Machine (KVM), a common feature in cloud environments such as Google Colab. On the other hand, the NVIDIA L4 GPU, with 24GB of VRAM, was configured to work in P8 mode, which is a low power consumption mode, with 12W/72W, i.e., with the current power usage (12 Watts) in relation to the card's maximum power capacity (72 Watts), and with CUDA 12.4 support.

Table 5
Group of tested cities from NSRDB data.

Number	Location	Latitude	Longitude	Climate
1	Hilo, Hawaii, USA	19.7070	-155.0810	Af
2	Iquitos, Loreto, Peru	-3.7437	-73.2516	Af
3	Manaus, Amazonas, Brazil	-3.1190	-60.0217	Af
4	Fort Lauderdale, Florida, USA	26.1224	-80.1373	Am
5	Miami, Florida, USA	25.7617	-80.1918	Am
6	Recife, Pernambuco, Brazil	-8.0476	-34.8770	Am
7	Aracaju, Sergipe, Brazil	-10.9472	-37.0731	As
8	Kailua-Kona, Hawaii, USA	19.6406	-155.9956	As
9	Kealahou, Hawaii, USA	19.5208	-155.9222	As
10	Laredo, Texas, USA	27.5306	-99.4803	BSh
11	Palm Springs, California, USA	33.8303	-116.5453	BSh
12	Tucson, Arizona, USA	32.2226	-110.9747	BSh
13	Albuquerque, N. Mex., USA	35.0844	-106.6504	BSk
14	Great Falls, Montana, USA	47.5002	-111.3008	BSk
15	Idaho Falls, Idaho, USA	43.4916	-112.0339	BSk
16	Lima, Lima, Peru	-12.0464	-77.0428	BWh
17	Needles, California, USA	34.8481	-114.6141	BWh
18	Phoenix, Arizona, USA	33.4484	-112.0740	BWh
19	Atlanta, Georgia, USA	33.7490	-84.3880	Cfa
20	Birmingham, Alabama, USA	33.5207	-86.8023	Cfa
21	Salt Lake City, Utah, USA	40.7608	-111.8910	Cfa
22	Craig, Alaska, USA	55.4761	-133.1483	Cfb
23	Prince Rupert, B.C., CAN	54.3150	-130.3208	Cfb
24	Vernon, B.C., CAN	50.2670	-119.2720	Cfb
25	Adak, Alaska, USA	51.8781	-176.6361	Cfc
26	Cold Bay, Alaska, USA	55.1853	-162.7219	Cfc
27	Unalaska, Alaska, USA	53.8898	-166.5426	Cfc
28	Modesto, California, USA	37.6391	-120.9969	Csa
29	Reno, Nevada, USA	39.5296	-119.8138	Csa
30	Sacramento, California, USA	38.5816	-121.4944	Csa
31	Astoria, Oregon, USA	46.1879	-123.8313	Csb
32	Eureka, California, USA	40.8021	-124.1637	Csb
33	Oakland, California, USA	37.8044	-122.2711	Csb
34	La paz, La paz, Bolivia	-16.5000	-68.1500	Cwb
35	Toluca, State of Mexico, Mexico	19.2826	-99.6557	Cwb
36	Wanchaq, Cusco, Peru	-13.5319	-71.9675	Cwb
37	Sioux Falls, South Dakota, USA	43.5460	-96.7313	Dfa
38	Montreal, Quebec, CAN	45.5017	-73.5673	Dfb
39	Saskatoon, Saskatchewan, CAN	52.1332	-106.6700	Dfb
40	Winnipeg, Manitoba, CAN	49.8956	-97.1386	Dfb

To investigate the robustness of the LLM and its rivals in long-term prediction, we standardize the evaluation process by using GHI univariate time series spanning 20 years (GHI data from 1998 to 2018 of 40 different locations from NSRDB, with a variety of climates). The dataset comprises an extensive, geographically diverse set of 40 cities, as listed in Table 5, distributed across the world, spanning several different latitudes. This collection covers a broad spectrum of Köppen climate classes, including fully humid equatorial regions (Af), monsoon-influenced tropical zones (Am), dry savanna climates (As), multiple arid and semi-arid types (BSh, BSk, BWh), warm and cold temperate subdivisions (Cfa, Cfb, Cfc, Csa, Csb, Cwb), and even cold continental climates (Dfa, Dfb). Such diversity ensures exposure to markedly different seasonal cycles, radiative regimes, humidity structures, and precipitation patterns.

We also listed each of these cities, in Fig. 3, and employed a zoom, illustrated in Fig. 4, to facilitate the reader visualization of details about such regions. This wide range of climatic contexts is essential for evaluating the robustness of the proposed approach. Models that succeed only in narrow environmental conditions may perform well locally, yet fail when applied to regions with distinct meteorological dynamics. By contrast, testing across climates that include monsoon-driven rainfall (e.g., Recife, Miami), strong diurnal thermal amplitudes in deserts (e.g., Phoenix, Needles), marine temperate environments (e.g., Prince Rupert, Eureka), and high-altitude Andean settings (e.g., La Paz, Cusco) imposes a stringent and comprehensive validation scenario. In addition to the 40 different locations, we used a years-long prediction horizon for long-term forecasting, following the horizon classifications presented in



Fig. 3. Illustration of the List of 40 cities utilized.

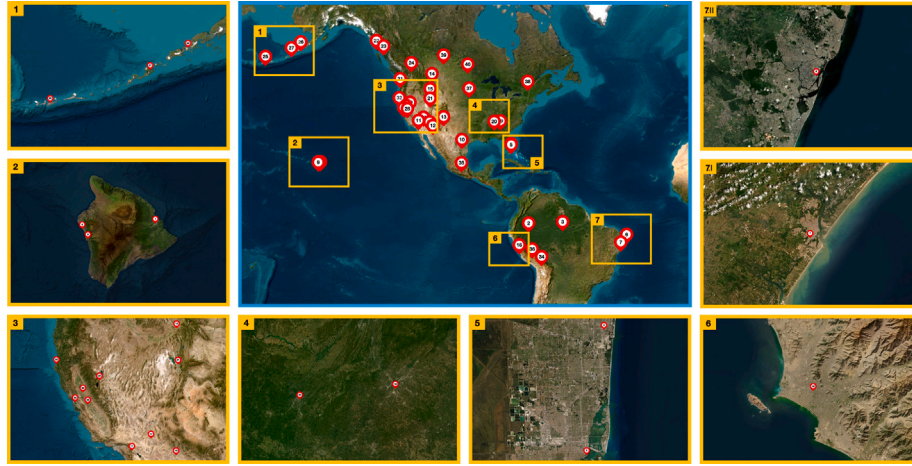


Fig. 4. Close-up of some areas that are considered difficult to see in Fig. 3.

reference [25]. More specifically, we set a 5-year horizon. We provide Algorithm 5 for reproducibility in this step.

In addition to algorithms 1-5, our experimental setup employs the same configurations for sampling resolution of the data collected from the NSRDB database (Δt), sampling resolution for long-term prediction ($\Delta \ell$), input length (N), prediction horizon (H), approach type (direct), loss function ($\mathcal{L}$), optimizer ($\mathcal{A}$), number of training epochs ($\mathcal{E}$), seed (ς), and learning rate (l_r). Table 6 lists all the hyperparameter values and parameter values used to ensure reproducibility. Alongside the data-driven methods Informer [48], Autoformer [49], TFT [50], PatchTFT [51], and Transformer [52], we also included strong domain baselines, known as Persistence [53], Smart Persistence [54], Quantized Smart Persistence (Q. Smart Persistence), and SARIMA [55], to assess the robustness of the proposed approach. It is important to mention that Smart Persistence is based on the clear-sky index and therefore uses the observed/measured global horizontal irradiance and the clear-sky global horizontal irradiance. However, the pipeline of the proposed method operates with 7-bit GHI quantization. Therefore, we implemented Q. Smart Persistence, which first computes the clear-sky index from physical irradiance values and then quantizes this index using the adopted quantization protocol. The quantized clear-sky index is used in the persistence step and is subsequently dequantized to reconstruct the predicted GHI. Furthermore, the SARIMA model was represented as $\text{SARIMA}(p, d, q)(P, D, Q)_s$, where p , d , and q denote, respectively, the autoregressive order, the differencing

order, and the moving-average order of the non-seasonal component, while P , D , and Q represent the corresponding orders of the seasonal component. In the implementation using `pmdarima.auto_arima`, the automatic search was performed with $p, q \leq 5$, $d \leq 2$, $P, Q \leq 2$, and $D \leq 1$, using the AIC criterion for model selection. Since the data have monthly resolution, the seasonal period was set to $s = 12$, corresponding to an annual cycle with 12 monthly samples.

To first illustrate the behavior of the evaluated models under contrasting climatic conditions, we report detailed results for two representative locations: Recife, Brazil, representing a tropical climate, and Reno, United States, representing a cold climate. Table 7 reports the forecasting performance obtained for Recife, Brazil, under the Am tropical monsoon climate. Among the domain baselines, Persistence obtained the weakest performance, with an MAE of 24.2500 W/m², an RMSE of 31.0915 W/m², a negative R^2 of -0.1744, a JSD of 0.4110, and an OVC of 0.3833. The strongest baseline performance was achieved by SARIMA in terms of MAE, RMSE, and JSD, with values of 14.2362 W/m², 18.3670 W/m², and 0.0747, respectively. Compared with these baselines, GPT-Neo achieved the best overall performance. It reduced the MAE to 12.4112 W/m² and the RMSE to 14.9128 W/m², while also obtaining the highest R^2 value of 0.7293 and the lowest JSD of 0.0105. The strongest result among the data-driven methods, excluding GPT-Neo, was achieved by Transformer in terms of MAE, RMSE, R^2 , and OVC, with values of 12.8269 W/m², 16.4880 W/m², 0.6691, and 0.9000, respectively.

Algorithm 5 Chronological splitting procedure for multi-city GHI data.

Require: Normalized monthly GHI time series $\hat{G}_i(t)$ for each location $i \in \{1, \dots, 40\}$

Ensure: Chronologically separated training, validation, and test subsets for each location

- 1: Define the sampling resolution:
- 2: 1 monthly averaged GHI sample (i.e., 1 sample/month, where each year contains 12 samples)
- 3: Define the forecasting horizon (H):
- 4: $H \leftarrow 5$ years ahead
- 5: Define the total length of each time series according to Python language
- 6: $T \leftarrow 20 \text{ years} \times 12 \text{ samples per year} = 240 \text{ samples}$, i.e., in Python we have $\{\hat{G}_i(t)\}_{t=0}^{239}$
- 7: **for all** locations $i \in \{1, \dots, 40\}$ **do**
- 8: Select the normalized GHI time series $D_i^{(T)} = \{\hat{G}_i(t)\}_{t=0}^{239}$
- 9: Split $D_i^{(T)}$ chronologically into three subsets:
- 10: $D_i^{(t)} \leftarrow$ first 10 years $\times 12$ samples per year = 120 samples, i.e., $D_i^{(t)} = \{\hat{G}_i(t)\}_{t=0}^{119}$
- 11: $D_i^{(v)} \leftarrow$ next 5 years $\times 12$ samples per year = 60 samples, i.e., $D_i^{(v)} = \{\hat{G}_i(t)\}_{t=120}^{179}$
- 12: $D_i^{(e)} \leftarrow$ final H years $\times 12$ samples per year = 60 samples, i.e., $D_i^{(e)} = \{\hat{G}_i(t)\}_{t=180}^{239}$
- 13: Store $D_i^{(t)}$, $D_i^{(v)}$, and $D_i^{(e)}$ for model training, validation, and testing
- 14: **end for**
- 15: Return $\{D_i^{(t)}, D_i^{(v)}, D_i^{(e)}\}_{i=1}^{40}$

Table 6

Experimental settings and hyperparameters for long-term prediction.

Parameters and Hyperparameters Θ	Configuration
Sampling resolution for forecasting (Δt)	1 sample/month (monthly average)
Input length (N)	60 samples
Forecasting horizon (H)	60 samples (5 years ahead)
Approach type	Direct
ADC resolution (b)	7 bits
Quantization levels (L)	128 levels
Normalization method	Min-max normalization
Normalization range	[0, 1]
Loss function ($\mathcal{L}$)	SMAPE
Optimizer ($\mathcal{A}$)	Adam
Training epochs ($\mathcal{E}$)	500
Learning rate ($\mathcal{L}_r$)	0.0001
Training batch size (β_{tr})	32
Validation batch size (β_{va})	32
Seed (ς)	1
Dropout rate (n_{dr})	10%
Heads in the reprogramming layer (n_{re})	8
Hidden layers in the LLM backbone (n_{hi})	32

However, TFT obtained the lowest JSD among the data-driven methods excluding GPT-Neo, with 0.0187, indicating a closer distributional match than Transformer, PatchTFT, Autoformer, and Informer. Compared with these data-driven methods, GPT-Neo achieved the best overall performance. It reduced the MAE to 12.4112 W/m² and the RMSE to 14.9128 W/m², while also obtaining the highest R^2 value of 0.7293 and the lowest JSD of 0.0105. Therefore, the results indicate that GPT-Neo is the most effective forecasting method in this scenario.

Table 8 reports the forecasting performance obtained for Reno, USA, under the Csa climate. Among the baselines, Persistence obtained the weakest performance, with an MAE of 38.7333 W/m², an RMSE of 47.7790 W/m², a negative R^2 of -0.6093, a JSD of 0.5685, and an OVC of 0.2333. This indicates that simple persistence was not suitable for representing the GHI dynamics in Reno. The strongest baseline was SARIMA,

Table 7

Recife, Pernambuco, Brazil (Am climate).

Models	MAE	RMSE	R^2	JSD	OVC
Persistence	24.2500 W/m ²	31.0915 W/m ²	-0.1744	0.4110	0.3833
Q. Smart Persistence	16.5641 W/m ²	20.6273 W/m ²	0.6041	0.0888	0.7667
Smart Persistence	16.5622 W/m ²	20.6477 W/m ²	0.6034	0.0887	0.7666
SARIMA	14.2362 W/m ²	18.3670 W/m ²	0.5901	0.0747	0.7500
Informer	22.1104 W/m ²	30.4329 W/m ²	-0.1273	0.0954	0.7167
Autoformer	15.4275 W/m ²	19.5389 W/m ²	0.5353	0.0398	0.8833
TFT	14.0855 W/m ²	17.8703 W/m ²	0.6113	0.0187	0.8667
PatchTFT	13.8226 W/m ²	17.4266 W/m ²	0.6303	0.0220	0.8500
Transformer	12.8269 W/m ²	16.4880 W/m ²	0.6691	0.0229	0.9000
GPT-Neo	12.4112 W/m ²	14.9128 W/m ²	0.7293	0.0105	0.8833

Table 8

City of Reno, Nevada, USA (Csa climate).

Models	MAE	RMSE	R^2	JSD	OVC
Persistence	38.7333 W/m ²	47.7790 W/m ²	-0.6093	0.5685	0.2333
Q. Smart Persistence	16.3890 W/m ²	21.4712 W/m ²	0.9392	0.0218	0.8394
Smart Persistence	16.3338 W/m ²	21.4389 W/m ²	0.9393	0.0218	0.8393
SARIMA	6.9833 W/m ²	9.2924 W/m ²	0.9391	0.0139	0.9000
Informer	23.7922 W/m ²	31.2473 W/m ²	0.3117	0.1598	0.5877
TFT	11.6761 W/m ²	14.3511 W/m ²	0.8548	0.0108	0.9000
Transformer	10.5909 W/m ²	12.9823 W/m ²	0.8812	0.0122	0.8889
Autoformer	9.5731 W/m ²	12.4765 W/m ²	0.8903	0.0226	0.8588
PatchTFT	7.6980 W/m ²	10.8258 W/m ²	0.9174	0.0361	0.8161
GPT-Neo	6.6909 W/m ²	8.3793 W/m ²	0.9505	0.0058	0.9167

which achieved the lowest MAE, RMSE, and JSD among the baselines, with 6.9833 W/m², 9.2924 W/m², and 0.0139, respectively. Compared with the baselines, GPT-Neo achieved the best overall performance. It obtained the lowest MAE and RMSE, 6.6909 W/m² and 8.3793 W/m², respectively, outperforming SARIMA, which was the strongest baseline. GPT-Neo also achieved the highest R^2 value, 0.9505, the lowest JSD, 0.0058, and the highest OVC, 0.9167. Regarding the data-driven methods, Informer obtained the weakest performance, with an MAE of 23.7922 W/m², an RMSE of 31.2473 W/m², an R^2 of 0.3117, a JSD of 0.1598, and an OVC of 0.5877. The strongest result among the data-driven methods, excluding GPT-Neo, was achieved by PatchTFT in terms of MAE, RMSE, and R^2 , with values of 7.6980 W/m², 10.8258 W/m², and 0.9174, respectively. However, TFT achieved the lowest JSD among the data-driven methods excluding the GPT-Neo, with 0.0108, and highest OVC, with 0.9000. Compared with these data-driven methods, GPT-Neo also showed superior performance. It improved upon PatchTFT, the strongest data-driven method excluding GPT-Neo in terms of MAE, RMSE, and R^2 , reducing the MAE from 7.6980 W/m² to 6.6909 W/m² and the RMSE from 10.8258 W/m² to 8.3793 W/m². In addition, GPT-Neo achieved a higher R^2 than PatchTFT, 0.9505 against 0.9174, and a lower JSD than TFT, 0.0058 against 0.0108, confirming its stronger overall forecasting capability.

Table 9 summarizes the average performance of all evaluated models across 40 distinct locations, offering a broader perspective on their generalization capabilities under diverse climatic and geographic conditions. Among the baselines, Persistence obtained the weakest performance, with an MAE of 34.2279 W/m², an RMSE of 41.3782 W/m², a negative R^2 of -0.5619, a JSD of 0.6039, and an OVC of 0.2085. The strongest baseline in terms of MAE and RMSE was SARIMA, with values of 13.0696 W/m² and 16.1269 W/m², respectively. However, Smart Persistence achieved the highest R^2 among the baselines, with 0.6859, as well as the lowest JSD, with 0.0638, and the highest OVC, with 0.8062. Regarding the data-driven methods, Informer obtained the weakest performance, with an MAE of 22.3302 W/m², an RMSE of 28.2847 W/m², an R^2 of 0.2452, a JSD of 0.1318, and an OVC of 0.6544. The strongest result among the data-driven methods, excluding GPT-Neo, was achieved by PatchTFT in terms of MAE, RMSE, R^2 , JSD, and OVC, with values of 9.5837 W/m², 12.2315 W/m², 0.7459, 0.0350, and

Table 9

Average results employing all 40 different cities.

Models	MAE	RMSE	R ²	JSD	OVC
Persistence	34.2279 W/m ²	41.3782 W/m ²	-0.5619	0.6039	0.2085
Q. Smart Persistence	15.9075 W/m ²	20.6551 W/m ²	0.6853	0.0665	0.8018
Smart Persistence	15.8989 W/m ²	20.6451 W/m ²	0.6859	0.0638	0.8062
SARIMA	13.0696 W/m ²	16.1269 W/m ²	0.5467	0.1401	0.7298
Informer	22.3302 W/m ²	28.2847 W/m ²	0.2452	0.1318	0.6544
Autoformer	11.0955 W/m ²	14.1500 W/m ²	0.7100	0.0374	0.8525
Transformer	11.0941 W/m ²	14.3278 W/m ²	0.7123	0.0432	0.8554
TFT	10.2210 W/m ²	12.8636 W/m ²	0.7350	0.0481	0.8477
PatchTFT	9.5837 W/m ²	12.2315 W/m ²	0.7459	0.0350	0.8680
GPT-Neo	9.4225 W/m ²	11.8149 W/m ²	0.7787	0.0378	0.8570

0.8680, respectively. Compared with the baselines, GPT-Neo achieved the best error performance, reducing the MAE to 9.4225 W/m² and the RMSE to 11.8149 W/m². It also obtained the highest R² value, 0.7787, outperforming SARIMA and the remaining baselines. Compared with the data-driven methods, GPT-Neo improved upon PatchTFT in MAE, RMSE, and R². However, PatchTFT achieved slightly better distributional metrics, with a lower JSD of 0.0350 against 0.0378 for GPT-Neo, and a higher OVC of 0.8680 against 0.8570. Even so, GPT-Neo provided the best overall forecasting accuracy across the 40-city evaluation.

Therefore, the results in Table 9 show that GPT-Neo performs well across geographically diverse conditions. PatchTFT emerges as a robust competitor. Nevertheless, GPT-Neo's combined superiority and balanced performance across all other metrics highlight the advantages of LLM-based forecasting as a general-purpose solution for multi-region climate prediction. Finally, Fig. 5 compares the observed test data (yellow curve) with the model predictions (blue curve) for Reno, using the five best-performing models according to the average results in Table 9. The results show that GPT-Neo provides a good visual agreement with the observed GHI curve, reinforcing its predictive consistency.

In addition to evaluating the robustness of the proposed method using MAE, RMSE, R², JSD, and OVC, we also report the average training

Table 10

Average training time across all 40 different cities.

GPT-Neo	Autoformer	TFT	Transformer	Informer	PatchTFT
188.9835 s	177.9763 s	80.8833 s	66.6192 s	62.4275 s	11.2982 s

Table 11

Average testing time using all 40 different cities.

GPT-Neo	Autoformer	Informer	TFT	Transformer	PatchTFT
0.6816 s	0.0955 s	0.0945 s	0.0920 s	0.0919 s	0.0875 s

Table 12

Number of parameters for each model.

Parameters	GPT-Neo	TFT	PatchTFT	Informer	Transformer	Autoformer
Trainable	53.9M	870K	454K	341K	292K	290K
Non-Trainable	125M	0	0	0	0	0
Total	≈ 179M	870K	454K	341K	292K	290K

and testing times across the 40 cities for the data-driven models, since these methods involve high computational demand. Tables 10 and 11 list the average training and testing times. Table 10 shows that GPT-Neo has the longest average training time (188.9835 s). However, this average training time is an expected result, since GPT-Neo is the most complex architecture in this manuscript, with 53.9M trainable parameters and 125M non-trainable parameters, as shown in Table 12 (≈ 179M total parameters), and based on the GPT-3 architecture. The second-longest average training time was obtained by Autoformer (177.9763 s), followed by TFT (80.8833 s), Transformer (66.6192 s), Informer (62.4275 s), and PatchTFT (11.2982 s), respectively. The results in Table 11 indicate that GPT-Neo again exhibits the highest average testing time (0.6816 s), which is also consistent with its large-scale architecture and

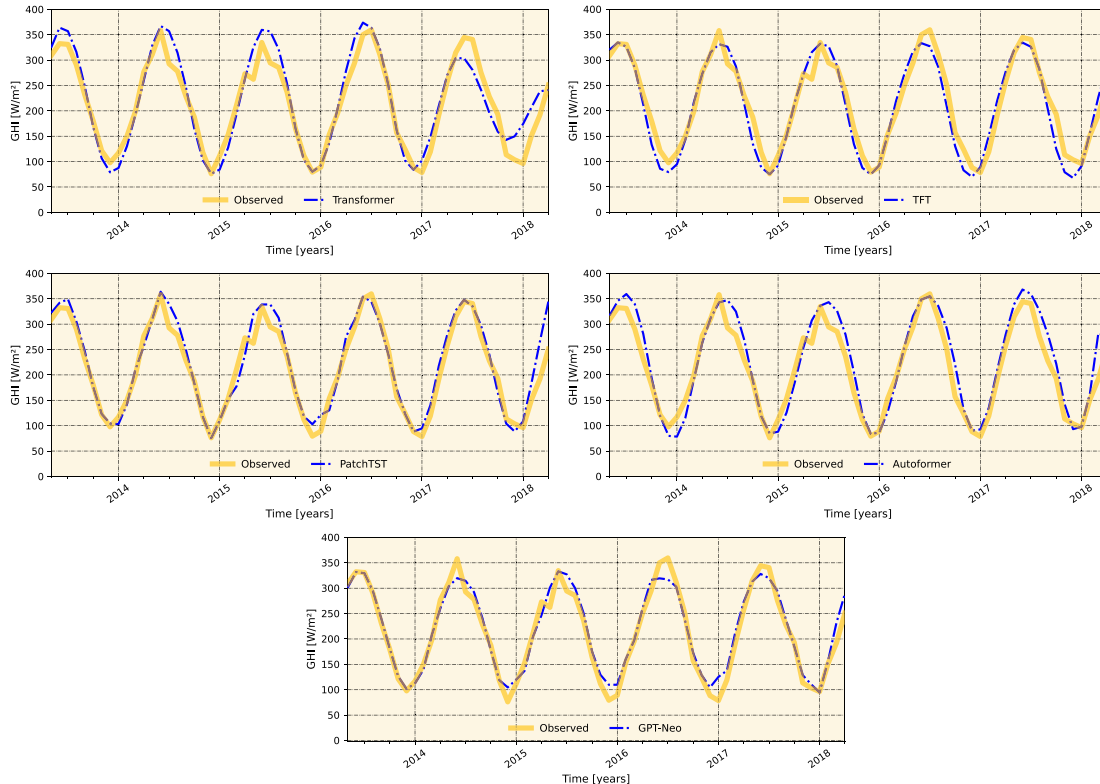
**Fig. 5.** Visual comparison of forecasting results for annual prediction (5-year horizon).

Table 13
Experimental settings and hyperparameters for daily prediction.

Parameters and Hyperparameters Θ	Configuration
Sampling resolution for forecasting ($\Delta\ell$)	1 sample/hour (hourly average)
Input length (N)	120 samples
Forecasting horizon (H)	120 samples (5 days ahead)
Approach type	Direct
ADC resolution (b)	7 bits
Quantization levels (L)	128 levels
Normalization method	Min-max normalization
Normalization range	[0, 1]
Loss function ($\mathcal{L}$)	MAE
Optimizer ($\mathcal{A}$)	Adam
Training epochs ($\mathcal{E}$)	200
Learning rate (l_r)	0.0001
Training batch size (β_{tr})	32
Validation batch size (β_{va})	32
Seed (ζ)	1
Dropout rate (n_{dr})	10%
Heads in the reprogramming layer (n_{he})	16
Hidden layers in the LLM backbone (n_{hi})	32

Table 14
Manaus, Amazonas, Brazil (Af climate).

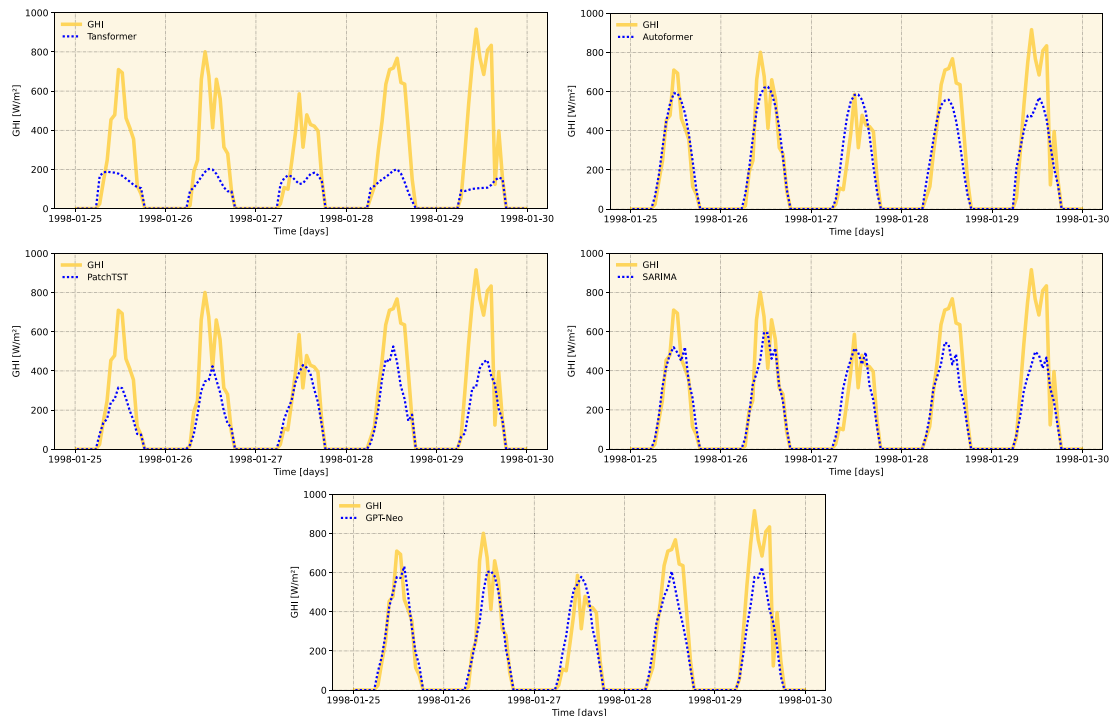
Models	MAE	RMSE	R ²	JSD	OVC
Smart Persistence	93.0952 W/m ²	170.5703 W/m ²	0.0847	0.2372	0.6583
Q. Smart Persistence	93.0643 W/m ²	170.4914 W/m ²	0.0847	0.1007	0.7992
Persistence	24.8333 W/m ²	41.2415 W/m ²	-0.5688	0.2132	0.6333
SARIMA	10.1995 W/m ²	15.3913 W/m ²	0.7815	0.0736	0.8416
TFT	24.9663 W/m ²	37.1591 W/m ²	-0.2736	0.2132	0.6333
Informer	22.2638 W/m ²	33.3120 W/m ²	-0.0235	0.1070	0.6981
Transformer	21.2299 W/m ²	31.8123 W/m ²	0.0665	0.1668	0.6859
PatchTFT	20.6456 W/m ²	25.2171 W/m ²	0.4135	0.1759	0.6750
Autoformer	10.3465 W/m ²	15.0053 W/m ²	0.7923	0.0655	0.7607
GPT-Neo	9.9846 W/m ²	14.1027 W/m ²	0.8165	0.0376	0.9000

high number of parameters. All remaining models exhibit lower testing times: Autoformer (0.0955 s), Informer (0.0945 s), TFT (0.0920 s), Transformer (0.0919 s), and PatchTFT (0.0875 s).

Besides, to strengthen the generalization/robustness evidence and further assess the models' ability to represent true irradiance dynamics, we introduce an additional daily case study based on hourly GHI data. This analysis complements the previous assessment, which examined performance under different climatic regimes, such as Recife and Reno, and across a wide, diverse set of 40 locations. In this case, we examine how the models perform under challenging, highly variable conditions, going beyond monthly averages by highlighting failure cases and presenting specific examples of complex days with high GHI variability. Thus, we tested the model under a scenario with rapid fluctuations and irregular patterns. For this purpose, we selected Manaus, Brazil (listed in Table 5), in January as a representative case of the rainy season. We selected January because it falls within the central Amazonian wet season, as defined by [56] as extending from December to April. This period of the year is a suitable case study for evaluating model behavior under more challenging and variable GHI conditions.

For this scenario, we also employed Algorithms 1–5, adapting them to the daily forecasting setting with hourly resolution. Here, the Algorithm 5 employs $T = 27$ days, comprising 22 days for training and 5 days for testing; therefore, $H = 5$ days. According to Table 13, our experimental setup employs the same configurations for sampling resolution of the data collected from the NSRDB database (Δt), sampling resolution for long-term prediction ($\Delta\ell$), input length (N), prediction horizon (H), approach type (direct), loss function ($\mathcal{L}$), optimizer ($\mathcal{A}$), number of training epochs ($\mathcal{E}$), seed (ζ), and learning rate (l_r). We also employed the data-driven architectures, comprising Transformer, Informer, Autoformer, TFT, PatchTFT, and GPT-Neo. For the GPT-Neo model, we updated the prompt prefix to reflect the new temporal resolution and forecasting horizon as follows: "The dataset contains data on hourly global horizontal irradiance. I need to predict five days ahead." In addition, we applied the same baseline models, such as Persistence, Smart Persistence, Q. Smart Persistence, and SARIMA, to this case study, adjusting the seasonality settings to the hourly resolution; i.e., for SARIMA, we set s to 24.

Table 14 reports the forecasting performance obtained for Manaus, Brazil, under the Af climate. Among the baselines, the strongest

**Fig. 6.** Visual comparison of forecasting results for daily prediction (5-day horizon).

approach was SARIMA, which achieved the lowest MAE, RMSE, and JSD among the baselines, with values of 10.1995 W/m², 15.3913 W/m², and 0.0736, respectively. Compared with the baselines, GPT-Neo achieved the best overall performance. It reduced the MAE to 9.9846 W/m² and the RMSE to 14.1027 W/m², outperforming SARIMA in terms of error magnitude. Regarding the data-driven methods, TFT obtained the weakest performance, with an MAE of 24.9663 W/m², an RMSE of 37.1591 W/m², a negative R^2 of -0.2736, a JSD of 0.2132, and an OVC of 0.6333. The strongest result among the data-driven methods, excluding the GPT-Neo, was achieved by Autoformer, with the lowest MAE, RMSE, and JSD, equal to 10.3465 W/m², 15.0053 W/m², and 0.0655, respectively. Compared with these data-driven methods, GPT-Neo again achieved the best overall performance. It improved upon Autoformer by reducing the MAE from 10.3465 W/m² to 9.9846 W/m² and the RMSE from 15.0053 W/m² to 14.1027 W/m². In addition, GPT-Neo achieved a higher R^2 than Autoformer, 0.8165 against 0.7923, as well as a considerably lower JSD, 0.0376 against 0.0655, confirming its promising capability to forecast GHI time series.

Therefore, the results in Table 14 show that GPT-Neo performs well under highly variable rainy-season conditions in Manaus. Autoformer emerges as the strongest competitor among the data-driven methods excluding GPT-Neo, while SARIMA remains the most competitive baseline in terms of MAE and RMSE. Nevertheless, GPT-Neo achieved the best overall balance across all evaluation metrics, including MAE, RMSE, R^2 , JSD, and OVC, indicating the advantages of LLM-based forecasting for capturing temporal dynamics under hourly GHI conditions. Finally, Fig. 6 compares the observed test data (yellow curve) with the model predictions (blue curve) for the 5-day-ahead hourly forecasting scenario in Manaus.

6. Conclusions

This work contributes to the field of long-term GHI prediction by introducing a pioneering LLM-based forecasting system built around the GPT-Neo architecture. The proposed method demonstrates that reliable long-term forecasting does not require highly complex architectures or large sets of input variables; instead, an LLM operating solely on GHI data can achieve competitive and superior performance. Our extensive evaluation across tropical, Mediterranean, and mixed climatic conditions confirms GPT-Neo's strong predictive capability, achieving the best results in Recife (a challenging case for all models) and Reno (a favorable case for all models), and maintaining robust performance across 40 additional locations. These results underscore the generalization and stability of LLM-based forecasting, surpassing traditional Transformer variants and offering a promising direction for future research. In addition, it is relevant to cite that experimentally evaluating LLMs' ability to incorporate contextual information and its effects on solar irradiance prediction represents a promising avenue for future research, as such an investigation requires a more in-depth analysis. It is worth mentioning that the GPT-Neo architecture exhibits a relatively long average training time; however, this is an expected result, as GPT-Neo is the most complex architecture in this manuscript, with 179M parameters, and is based on the GPT-3 architecture. However, we demonstrated that the lowest average training cost per trainable parameter is that of GPT-Neo (3.51×10^{-6} s, as demonstrated in Table 12). Furthermore, the comprehensive literature review conducted in this study highlights persistent gaps in current methodologies, particularly the absence of context-aware models and scalable long-term forecasting solutions. By addressing these gaps, GPT-Neo establishes a new benchmark for LLM-driven GHI prediction and paves the way for more accurate, efficient, and practical forecasting systems. The insights presented here are directly relevant to academic research and hold significant potential for renewable energy applications, particularly in supporting photovoltaic integration, energy planning, and climate-aware decision-making.

CRedit authorship contribution statement

Thales W. Cabral: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Gustavo Fraidenraich:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Miriam A.M. Capretz:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data are public and available on the National Solar Radiation Database (NSRDB) site.

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